

Homework Assignment 1 (9/15/2013)

46 Points — Due Fr 9/27/2013 (per e-mail by 3pm)

Please submit the answers to this homework assignment per e-mail by 3pm on Fr 9/27/2013. I need the following files:

- (i) A single R file with all your R commands used to answer this homework assignment, named `lastname_firstname_HW01.R` (so I can verify your results).
- (ii) Data files, named `lastname_firstname_HW01_data1.EXT`, ..., `lastname_firstname_HW01_dataN.EXT` where EXT may be any valid data format, such as `tex`, `csv`, `xlsx`, etc. If you read your data directly from a web page or if you use a data set from R, then there is no need to submit that data set.
- (iii) A single pdf file that contains the R code, your results (maps, numerical output, etc.), and any explanatory text for each question. If you (already) know how to use Sweave or any other tool that automatically combines R output with text documents, you can certainly use this.

Questions:

- (i) (10 Points) Create (at least) two meaningful choropleth maps for the United States, using the R package `maps`.

For one of these maps, break the data down into quartiles, i.e., for each class, there should be either 12 or 13 states that appear in the same color. Use a sequential color scheme. For the other map, break the data down into five equally wide intervals. Check your intervals and avoid that there are intervals with no observations at all. However, it would be fine that some of these intervals contain only one observation (e.g., the maximum and/or the minimum). Use a divergent color scheme. To obtain these color schemes, start at <http://colorbrewer2.org/> and continue with the R package `RColorBrewer` once you have decided on a color scheme. Compare your two (or more) maps and discuss the visual impact of each of these.

You can work with a data set of your own choice. If you can't find anything suitable, use one of the `state.x77` data sets.

- (ii) (10 Points) Create a simple linked micromap plot for the United States, either using the R package `micromap` or `micromapST`.

Work with the same data set as in question (i). Compare your micromap with your two (or more) plots from question (i) and comment on the advantages and disadvantages of each of these three (or more) plots.

- (iii) (10 Points) Create a meaningful google map overlay, using the R package `RgoogleMaps`. Decide whether a satellite map or a road map is more suitable for your data. Often, a grayscale map is of advantage.

You can work with a data set of your own choice. If you can't find anything suitable, use one of the data sets from the R package `splancs`.

If your data set contains a categorical variable (such as species or gender), use that variable to distinguish between the different categories (e.g., color locations for men in blue and locations for women in red).

Describe the overall pattern you see. Is there any noticeable difference for the various categories?

- (iv) (16 Points) **Note: This question originates from Quizzes 2 and 3 of the “Introduction to R” class from Spring 2013. Once this class has ended on Wed 9/25/2013, you should have all necessary knowledge to answer this question. Keep in mind that this originally was a closed notes/closed computer question! Therefore, I would suggest that you first answer this question on paper and only check your answers on the computer at the end.**

When dealing with earthquakes, we are usually interested in their locations, and not only in their magnitudes. In this question, we will focus on earthquakes that occurred in the San Francisco area.

Below are a few expressions and results that should provide you with some useful background information on the structure of the earthquake data:

```
CAquakes = read.table(file = "http://www.consrv.ca.gov/cgs/rghm/quakes/Documents/ms49epicenters.txt",
                      header = TRUE)

> dim(CAquakes)
[1] 383  4
>
> class(CAquakes)
[1] "data.frame"
>
> class(CAquakes$Latitude)
[1] "numeric"
>
> class(CAquakes$Longitude)
[1] "numeric"
>
> head(CAquakes)
  Date Latitude Longitude  M
```

```

1 18001011    36.8    -121.5 5.5
2 18001122    32.9    -117.8 6.3
3 18030000    34.2    -118.1 5.5
4 18030525    32.8    -117.1 5.5
5 18060325    34.4    -119.7 5.5
6 18080621    37.8    -122.6 5.5
>
> tail(CAquakes)
      Date Latitude Longitude  M
378 19951228  38.7210 -119.6590 5.5
379 19970122  40.2720 -124.3940 5.6
380 19981127  40.6670 -125.3840 5.6
381 19991016  34.6815 -116.2848 5.8
382 19991016  34.6000 -116.2700 7.1
383 19991016  34.4400 -116.2500 5.7

```

The map (see Figure 1) has been created in R with the following commands:

```

library(RgoogleMaps)

bb = qbbox(c(min(CAquakes$Latitude), max(CAquakes$Latitude)), c(min(CAquakes$Longitude), max(CAquakes$Longitude)))

MyMap = GetMap.bbox(bb$lonR, bb$latR, destfile = "CAquakes.png", GRAYSCALE = TRUE)

tmp = PlotOnStaticMap(MyMap, lat = CAquakes$Latitude, lon = CAquakes$Longitude,
  cex = 1.5, pch = 20, col = "red", add = FALSE, axes = TRUE, mar = c(3, 3, 0, 0))

tmp = PlotOnStaticMap(MyMap, lat = c(36, 36, 39, 39, 36), lon = c(-124, -120, -120, -124, -124),
  col = "black", add = TRUE, FUN = lines, lwd = 2)

```

For this question, assume that we are primarily interested in earthquakes that occurred in the San Francisco area. This is the rectangular area (strictly) inside the black box in Figure 1. The lower left corner for this area is (Latitude, Longitude) = (36, -124) and the upper right corner is (Latitude, Longitude) = (39, -120). Recall that negative values for longitude represent locations west of the Greenwich Meridian, i.e., a longitude of -120 alternatively can be written as 120°W.

- (a) First, create a logical vector (of length 383), called `inSFarea`, that contains the information whether the epicenter of an earthquake was (strictly) inside the area specified above, i.e., it is `TRUE` if (Latitude, Longitude) is (strictly) inside this area and `FALSE` otherwise. Be careful with the signs and the directions of inequalities! Use this vector as part of your answer in the following parts whenever possible.

`inSFarea =`

Now, indicate R expressions to answer the following questions. As usual, be as efficient as possible.

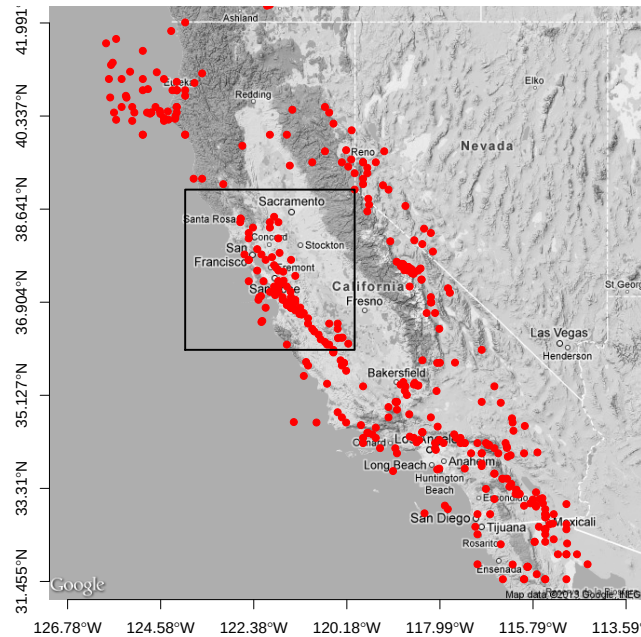


Figure 1: Location of earthquakes in California from 1800 to 1999. The area of interest around San Francisco is inside the rectangular area.

- (b) How many earthquakes were in the area of interest?
- (c) What was the average (= mean) magnitude of the earthquakes in the area of interest?
- (d) Report the dates of the earthquakes in the area of interest since January 1, 1990.
- (e) What was the highest magnitude of an earthquake in the area of interest from January 1, 1900, to December 31, 1910 — and when did it occur? You may want to use more than one expression here (but you still need to be as efficient as possible!).

Use a function from the **apply** family in **all** of your answers below (this can be a different one for each answer).

- (f) What was the median magnitude in the area of interest and what was the median magnitude outside the area of interest? The result should be a vector of length 2.
- (g) Someone is interested in the geographic center of all the earthquakes recorded in this data file. You may disagree that this is a good measure to summarize the spatial locations, but nevertheless, this person insists that you calculate this center point. So, what was the average (= mean) latitude and the average (= mean) longitude of **all** the earthquakes? The result should be a vector of length 2.
- (h) How many of the recorded 383 earthquakes were recorded in the 19th century (1800–1899) and how many were recorded in the 20th century (1900–1999)? The result should be a vector of length 2.